



Which are the critical factors that influence investment decisions on the energy transition?

Fields marked with * are mandatory.

1 Consent Form

Background

Energy system models (ESMs) have been used by experts to project and describe the long-term impacts of interventions on energy systems. While ESMs have yielded useful results in simulating energy supply, there is a lack of accuracy in simulating energy consumption, especially in the residential sector, where diversity of dwellings and human behaviours produce a broad spectrum of consumption patterns. To address this issue, the EU-funded project WHY will quantitatively analyse people's daily decisions regarding energy consumption and their reactions to specific policy-driven interventions. Furthermore, the project will examine how these daily behaviours translate into electricity consumption with the help of causal modelling. In the context of the project, interventions are external stimuli that have an impact on private settings such as changes in funding schemes, changes in policies, changes in taxes or subsidies, etc.

What do we expect from you?

In the framework of the H2020-WHY (why-h2020.eu), H2020-PARITY (parity-h2020.eu/), H2020-SMARTLIVINGEPC (cordis.europa.eu/project/id/101069639) and H2020-ATELIER (smartcity-atelier.eu) projects, we are assessing the determinants (factors) that lead consumers to participate in the energy transition in four different aspects: flexibility markets (e.g. PV panels), energy efficiency (e.g. building insulation), mobility (e.g. electric vehicles) and energy conservation (e.g. sharing economy). Our objective is to build a set of archetypes (a very typical example of a certain person which is present in the collective unconscious) that characterise as much as possible the European population in order to create better

policies for them. To this end, your core contribution to this research consists of self-assessing how much you agree with a set of 32 statements.

Rights as Stakeholder

- Your participation in this project is voluntary and no costs are derived for you. You have the right to withdraw at any point during the activity, for any reason, and without any prejudice.
- No personal data will be collected.
- All your contributions will be used only for the sole purpose of research (e.g. dissemination, outreach, etc.).
- Data contained in this survey will be kept at least until 5 years after the end of the project for auditing or reporting purposes by request of the funding agencies. A copy of your answers could be collected at the end of the survey following the steps provided by the system.
- The researchers in charge of this activity are Cruz E. Borges (cruz.borges@deusto.es) and Diego Casado-Mansilla (dcasado@deusto.es). You can contact them to solve any question related to this survey.

Consent Form

By hitting the next button, you acknowledge that:

- Your participation in the action is voluntary.
- You are at least 18 years of age.
- You have read the background information that provides enough details about the project (purpose, expected duration and procedures of the study).
- You have been informed about your right to refuse to participate or to leave the activity at any moment without any justification.
- You have been notified of the contact persons, in the case you have questions or doubts during the activity.
- You have been informed how to get a copy of the consent form and your answers.
- You have had enough time to decide on your participation in the study.
- You have been informed about the questionnaire that you have been asked to complete.
- You have been informed about the storage procedures of the study data.
- You allow experts involved in the study under confidentiality agreements to utilize the information for the purpose of the study and only for this.

☐ I agree to participate in the study

2 Scenarios for energy transition

2.1 Rank in order of priority, from top to bottom, in which scenario you will be willing to live or to contribute /invest to make it possible. The higher up you rank a scenario, the more priority it has (the scenario at the top is the highest priority and the scenario at the bottom is the lowest). Please note that we understand that not all apply to you or that you may think they are futuristic, but please try to rank them according to your personal preference or situation.

Use drag&drop or the up/down buttons to change the order or accept the initial order.



Scenario a. One of the appliances you use occasionally (e.g. the oven for baking, the washing/drying machine, a drill or the vacuum cleaner) breaks down again. In order to contribute to the climate neutrality objectives, you decide not to buy a new piece of equipment for individual use but to propose its purchase to your community for shared and common use (and that way you can buy the highest energy efficiency option). To do this, you convince your entire building community to share these household appliances for common everyday use. Luckily, you have space in the basement to accommodate them and you propose to your community to adhere to this initiative when they have to replace their equipment and not before.



Scenario b. You live in a new or highly refurbished passive house (https://en.wikipedia.org/wiki/Passive_house). In addition to having first-class facade, roof and window insulation, an intelligent and highly efficient heating/air conditioning system supplies the dwelling with your heating, cooling and hot water needs. An energy management system controls the heating/air conditioner system giving it the potential to share energy with the neighbors building a “distributed district heating”. Moreover, you are very environmentally conscious so you prioritize saving energy with any kind of investment of money or time, and you are always willing to sacrifice some comfort when setting the thermostat temperature.



Scenario c. You live in a new or highly refurbished passive house (https://en.wikipedia.org/wiki/Passive_house). You have solar panels integrated into the building and a battery system that allows you to store excess daily energy to consume at night. The generation system is slightly oversized, and in summer, you have a surplus of energy generated that is used at the community electrolyzer that generates and stores hydrogen. This hydrogen is later used in a highly efficient community cogeneration system (a.k.a. technology that produces electricity and thermal energy) to power a low-temperature district heating that provides the heating/cooling and hot water demand of the community. All these components could be controlled by an energy management system in which you can invest, not only to optimize its performance but also to allow you to trade energy between neighbors or with the energy provider.



Scenario d. The city where you live has been re-designed in a way that all services are at foot distance (for example, following the idea of 15 mins cities - https://en.wikipedia.org/wiki/15_minute_city). A series of agreements between the government and companies, aligned with investments in behavior change from the citizens’ side, ensures that teleworking is mandatory at least three days a week. In your city, the number of vehicles is drastically reduced due to several incentives to embrace low-emissions mobility vehicles (e.g., bikes, e-bikes, e-scooters, etc.) and facilities to take public transport. Indeed, a combination of public or private personal mobility, robotaxis (a.k.a. self-driving taxi or a driverless taxi) and electric public transport supply the rest of travel needs (inter and intra city). For these reasons, traffic jams are something from the past and the air is again breathable. Long distance transport of passengers and goods is only made using trains and the amount of travel by plane is drastically reduced to intercontinental travel.

Imaging that you have inherited a large amount of money or an institution has provided you a credit in very good conditions (e.g. 250 k€). How much money (in percentage) would you be willing to spend on each of the scenarios provided in the previous question? Please answer this question in the following 5 boxes (2.2 to 2.6) taking into consideration that there is one box (2.6) to be used if you do not want to invest all your money but to save some. Finally, bear in mind that you should allocate all the money in the 5 boxes, namely, the sum of all figures provided in boxes 2.2 to 2.6 below should add to 100.

2.2 Percentage spend for scenario a)

Only values of at most 100 are allowed

 %

2.3 Percentage spend for scenario b)

Only values of at most 100 are allowed

 %

2.4 Percentage spend for scenario c)

Only values of at most 100 are allowed

 %

2.5 Percentage spend for scenario d)

Only values of at most 100 are allowed

 %

2.6 Percentage saved

Only values of at most 100 are allowed

 %

2.7 Imaging that you have bought a new house that needs a complete refurbishment, how much money (in €) will you be willing to invest in improving the efficiency of the house (like including energy efficient appliances, improving the insulation, installing solar panels, purchasing an electric car or several of the previous ones together)?

Only values of at most 1000000 are allowed

 €

2.8 Now assume that the government gives you an interest free credit with a long term payment period for the total cost of the refurbishment. How much will you be willing to invest in that case?

Only values of at most 1000000 are allowed

 €

2.9 Assume that the supply of electricity suffers frequent but short interruptions in your neighborhood due to a bad grid infrastructure. This way, **only a small number of houses are affected and the general communication and service infrastructure is still working**. The utility provider has a partial solution to temporarily provide a limited local electricity supply that cannot support all the normal loads. So they are asking which services you would prefer in order to estimate the resulting loads. Please, rank the following energy services from 0 to 10 stars where 0 stars means it is extremely low priority for you and 10 stars is absolutely needed for you:

Ensure that drinking water is available in my home.	
Allow me to cook.	

Allow me to commute long distances.	
Allow me to communicate with my family and peers.	
Allow me to clean the house.	
Allow me to heat my house.	
Allow me to cool my house.	
Allow me to move goods.	
Allow me to work at home.	
Provide me and the family with entertainment.	
Allow me to light up my house.	
Allow me to operate a Home Energy Management system.	
Allow me to operate the washing machine, dishwasher, etc.	
Allow me to keep the food refrigerated.	
Allow me to generate hot water for the appliances.	
Allow me to generate hot water for showering and cleaning.	

Allow me to operate home security systems.	
Allow me to operate my smart home devices (automatic window openers, shutters, etc.)	
Allow me to operate ventilation.	

2.10 Assume that full black-outs could occur in your region. This mean **the entire region** (and possibly even beyond) **is without electricity supply for at least a day or two**. Please note that in this situation **several services are not working or working** only in a very limited way (like cellular network, internet, television, etc.). The utility provider has a partial solution to provide limited local electricity supply but cannot support the electricity supply all the normal loads so they are asking which loads they should prioritize. Please, rank the following energy services from 0 to 10 stars where 0 stars means it is extremely low priority for you and 10 stars is absolutely needed for you:

Ensure that drinking water is available in my home.	
Allow me to cook.	
Allow me to commute long distances.	
Allow me to communicate with my family and peers.	
Allow me to clean the house.	
Allow me to heat my house.	
Allow me to cool my house.	
Allow me to move goods.	

Allow me to work at home.	
Provide me and the family with entertainment.	
Allow me to light up my house.	
Allow me to operate a Home Energy Management system.	
Allow me to operate the washing machine, dishwasher, etc.	
Allow me to keep the food refrigerated.	
Allow me to generate hot water for the appliances.	
Allow me to generate hot water for showering and cleaning.	
Allow me to operate home security systems.	
Allow me to operate my smart home devices (automatic window openers, shutters, etc.)	
Allow me to operate ventilation.	

2.11 Select the statement that best represents yourself when you are taking a decision related to the **scenario you selected as the highest priority in the previous question**:

- ☐ b. I prefer to do things with the lowest physical and mental effort. I am concerned about the environment, but I am usually too busy to think twice before acting.
- ☐ c. I prefer to think carefully about my decisions overall if they impact the climate. I know what I want and take my time to evaluate my options.
- ☐ a. I prefer to not make a decision. I rather try to continue doing the same every time without thinking and prioritize following my daily routines.
- ☐ d. None of the above.
- ☐ e. I prefer not to answer.

2.12 Select the statement that best represents yourself when you are taking a decision related to the **scenario you selected as the highest priority in the previous question**:

- ☐ e. Society or my peers influence me, so I usually follow what others suggest to me.
- ☐ b. I am concerned about the environment, but I always prioritize comfort.
- ☐ f. I am indifferent to environmental concerns.
- ☐ h. I prefer not to answer.
- ☐ a. I am an environmentally conscious and a well-informed consumer.
- ☐ d. I am rather materialist and the impact on the environment of my decisions is beyond my personal responsibility.
- ☐ g. None of the above.
- ☐ c. I am concerned about the environment, but I lack awareness of my impacts to act differently.

2.13 Select the statement that best represents yourself when you are taking a decision related to the **scenario you selected as the highest priority in the previous question**:

- ☐ d. I'd only invest in the energy transition if it helps to improve my safety and that of my relatives.
- ☐ a. I consider myself as an early adopter, so I enjoy following the latest trends even if it means taking some risks or impacts on the environment.
- ☐ e. I'd only invest in the energy transition if it helps me to reduce my environmental impact in any form.
- ☐ g. I'd only invest in the energy transition if it helps to maximize my comfort/well-being and that of my relatives.
- ☐ c. My main interest is taking the most cost-effective decision. I am an informed person and I always make some calculations before making any decision.
- ☐ f. I'd only invest in the energy transition if it helps me to increase my social status (i.e., the opinion of others about me) or if it is the social norm of my colleagues, friends and family.
- ☐ i. I prefer not to answer.
- ☐ h. None of the above.
- ☐ b. I am not really interested in new equipment, technology or environmental actions, so I just use/do the one that I am used to.

3 Determinants (factors)

3.1.1 As you can see, these ideal yet futuristic scenarios presented in the previous section would only be possible if citizens, enterprises, and public authorities collaborate. In the following, you will be presented with a list of statements describing factors that may influence your decision making when it comes to contributing to make this scenario possible (e.g., change your daily behavior, take investment decisions with money or time or decide who to vote for to make the scenarios a reality). Please rate each of them, **based only on the situation you selected as the highest priority in the previous question**, in a scale being the left-side of the slider “Not at all important for you” and the right-side of the slider “Drastically important to you”.

3.1.1.1 a. Profits are what guide my decision making, I always prefer to earn or save money with every decision I take.

3.1.1.2 b. Access to funding (my own savings, deductions, exemptions, and/or credits) is the main factor that allows/hinders me to make an investment decision.

3.1.1.3 c. The evaluation of the risks of my investment(s) is what will always guide my final decision.

3.1.1.4 d. I will only invest if my actions have an impact beyond the monetary gain/losses.

3.1.1.5 e. I am a thrifty person, so I only invest in actions that allow me to reduce my cost/impact /expenditures.

3.1.1.6 f. Every decision I take serves to foster the planet's preservation. If my choice might harm the environment, I will always avoid taking this action.

3.1.1.7 g. Having complete certainty that my actions comply with the legal, tax and administrative regulations are what guide my actions.

3.1.1.8 h. I only make decisions if I trust all the parties involved (e.g., public administration, neighbors) and that I trust the technology that is needed to accomplish my goal.

3.1.1.9 i. I only make decisions if the outcome of them ensures or improves my safety or the ones of my relatives.

3.1.1.10 j. I always review and assess the pros and cons of my decisions looking for the most cost-effective option.

3.1.1.11 k. I do not make a decision if I do not have enough knowledge of the subject matter.

3.1.1.12 l. Feeling that I am competent to make an investment is what guides my decision making.

3.1.1.13 m. I carefully check that the technology or equipment fits my lifestyle or the technical requirements before making an investment decision.

3.1.1.14 n. I always review and assess the pros and cons of my decisions in relation to the environment before making a decision.

3.1.1.15 o. I will only make a decision if I feel satisfied with the action and the expected outcome.

3.1.1.16 p. I only make a decision if I feel personally committed to the action and the expected outcome.

3.1.1.17 q. I will only make a decision if I feel that I can sustain it throughout time.

3.1.1.18 r. Self-sufficiency and individual sovereignty is what guide my decisions. I will only make a decision if I feel that the investment will improve my control of all circumstances and potential outcomes.

3.1.1.19 s. I will only make a decision if it improves my well-being or the well-being of my relatives.

3.1.1.20 t. I will only make a decision if it improves my comfort or the comfort of my relatives.

3.1.1.21 u. I firmly believe that we live in a society where we have to adhere to regulations, laws and community agreements by all means, so my investment decision has to agree with this vision.

3.1.1.22 v. My investment decisions are influenced by the opinions of others (such as my peers, relatives or family).

3.1.1.23 w. I only make an investment decision if it fulfill a social need or improve the society as a whole.

3.1.1.24 x. I will only make an investment decision if it improves my possibilities to socialise with my peers and relatives.

3.1.1.25 y. I will only make an investment decision if the people affected by it (for example, my relatives, peers or the community) agree with the decision cohesively.

3.1.1.26 z. I love to test new ideas and cutting-edge technology so novelty is what drives my investment decisions.

3.1.1.27 aa. Having fun is important to me. Therefore, I will only make a decision if it would be enjoyable and amusing for me.

3.1.1.28 ab. I make investment decisions that lead to my increased status and I can show others what I achieved.

3.1.1.29 ac. I usually follow the trends when making a decision. In particular, I usually find myself sticking to the ads I see, the recommendations of people I admire, or what I read in magazines or blogs I follow.

3.1.1.30 ad. I only make a decision if it helps me improve my position as an expert on the subject matter.

3.1.1.31 ae. I only make a decision if it improves my peers' opinions about me, even if this decision is not always what I would do only for myself.

3.1.1.32 af. I only make a decision if the action has a personal, inner meaning for me (beyond any economic gain).

4 Socio-economic description

4.1 Have you understood so far the questions we have made?

- ☐ -1000 * 23 ^27
- ☐ It is an average survey.
- ☐ Car
- ☐ No, the survey is quite difficult to answer.
- ☐ 0
- ☐ Yes, it has been easy so far.

* 4.2 Do you have a Prolific ID? (if you don't know what this is, just answer no)

- ☐ Yes
- ☐ No

* 4.3 Please provide your Prolific ID

24 character(s) maximum

4.4 In which country do you reside?

- ☐ AF - Afghanistan
- ☐ AL - Albania
- ☐ DZ - Algeria
- ☐ AD - Andorra
- ☐ AO - Angola
- ☐ AG - Antigua and Barbuda
- ☐ AR - Argentina
- ☐ AM - Armenia
- ☐ AU - Australia
- ☐ AT - Austria
- ☐ AZ - Azerbaijan
- ☐ BS - Bahamas
- ☐ BH - Bahrain
- ☐ BD - Bangladesh
- ☐ BB - Barbados
- ☐ BY - Belarus
- ☐ BE - Belgium
- ☐ BZ - Belize
- ☐ BJ - Benin
- ☐ BT - Bhutan
- ☐ BO - Bolivia
- ☐ BA - Bosnia and Herzegovina
- ☐ BW - Botswana
- ☐ BR - Brazil
- ☐ BN - Brunei Darussalam
- ☐ BG - Bulgaria
- ☐ BF - Burkina Faso
- ☐ BI - Burundi
- ☐ CV - Cabo Verde
- ☐ KH - Cambodia
- ☐ CM - Cameroon
- ☐ CA - Canada
- ☐ CF - Central African Republic
- ☐ TD - Chad
- ☐ CL - Chile
- ☐ CN - China

- ☐ CO - Colombia
- ☐ KM - Comoros
- ☐ CG - Congo
- ☐ CR - Costa Rica
- ☐ CI - Côte D'Ivoire
- ☐ HR - Croatia
- ☐ CU - Cuba
- ☐ CY - Cyprus
- ☐ CZ - Czechia
- ☐ CD - Democratic Republic of the Congo
- ☐ DK - Denmark
- ☐ DJ - Djibouti
- ☐ DM - Dominica
- ☐ DO - Dominican Republic
- ☐ EC - Ecuador
- ☐ EG - Egypt
- ☐ SV - El Salvador
- ☐ GQ - Equatorial Guinea
- ☐ ER - Eritrea
- ☐ EE - Estonia
- ☐ SZ - Eswatini
- ☐ ET - Ethiopia
- ☐ FJ - Fiji
- ☐ FI - Finland
- ☐ FR - France
- ☐ GA - Gabon
- ☐ GM - Gambia
- ☐ GE - Georgia
- ☐ DE - Germany
- ☐ GH - Ghana
- ☐ GR - Greece
- ☐ GD - Grenada
- ☐ GT - Guatemala
- ☐ GN - Guinea
- ☐ GW - Guinea Bissau
- ☐ GY - Guyana
- ☐ HT - Haiti
- ☐ HN - Honduras
- ☐ HU - Hungary
- ☐ IS - Iceland
- ☐ IN - India
- ☐ ID - Indonesia
- ☐ IR - Iran
- ☐ IQ - Iraq
- ☐ IE - Ireland
- ☐ IL - Israel
- ☐ IT - Italy

- ☐ JM - Jamaica
- ☐ JP - Japan
- ☐ JO - Jordan
- ☐ KZ - Kazakhstan
- ☐ KE - Kenya
- ☐ KI - Kiribati
- ☐ KW - Kuwait
- ☐ KG - Kyrgyzstan
- ☐ LA - Laos
- ☐ LV - Latvia
- ☐ LB - Lebanon
- ☐ LS - Lesotho
- ☐ LR - Liberia
- ☐ LY - Libya
- ☐ LI - Liechtenstein
- ☐ LT - Lithuania
- ☐ LU - Luxembourg
- ☐ MG - Madagascar
- ☐ MW - Malawi
- ☐ MY - Malaysia
- ☐ MV - Maldives
- ☐ ML - Mali
- ☐ MT - Malta
- ☐ MH - Marshall Islands
- ☐ MR - Mauritania
- ☐ MU - Mauritius
- ☐ MX - Mexico
- ☐ FM - Micronesia
- ☐ MC - Monaco
- ☐ MN - Mongolia
- ☐ ME - Montenegro
- ☐ MA - Morocco
- ☐ MZ - Mozambique
- ☐ MM - Myanmar
- ☐ NA - Namibia
- ☐ NR - Nauru
- ☐ NP - Nepal
- ☐ NL - Netherlands
- ☐ NZ - New Zealand
- ☐ NI - Nicaragua
- ☐ NE - Niger
- ☐ NG - Nigeria
- ☐ KP - North Korea
- ☐ MK - North Macedonia
- ☐ NO - Norway
- ☐ OM - Oman
- ☐ PK - Pakistan

- ☐ PW - Palau
- ☐ PA - Panama
- ☐ PG - Papua New Guinea
- ☐ PY - Paraguay
- ☐ PE - Peru
- ☐ PH - Philippines
- ☐ PL - Poland
- ☐ PT - Portugal
- ☐ QA - Qatar
- ☐ MD - Republic of Moldova
- ☐ RO - Romania
- ☐ RU - Russian Federation
- ☐ RW - Rwanda
- ☐ KN - Saint Kitts and Nevis
- ☐ LC - Saint Lucia
- ☐ VC - Saint Vincent and the Grenadines
- ☐ WS - Samoa
- ☐ SM - San Marino
- ☐ ST - Sao Tome and Principe
- ☐ SA - Saudi Arabia
- ☐ SN - Senegal
- ☐ RS - Serbia
- ☐ SC - Seychelles
- ☐ SL - Sierra Leone
- ☐ SG - Singapore
- ☐ SK - Slovakia
- ☐ SI - Slovenia
- ☐ SB - Solomon Islands
- ☐ SO - Somalia
- ☐ ZA - South Africa
- ☐ KR - South Korea
- ☐ SS - South Sudan
- ☐ ES - Spain
- ☐ LK - Sri Lanka
- ☐ SD - Sudan
- ☐ SR - Suriname
- ☐ SE - Sweden
- ☐ CH - Switzerland
- ☐ SY - Syrian Arab Republic
- ☐ TJ - Tajikistan
- ☐ TZ - Tanzania
- ☐ TH - Thailand
- ☐ TL - Timor-Leste
- ☐ TG - Togo
- ☐ TO - Tonga
- ☐ TT - Trinidad and Tobago
- ☐ TN - Tunisia

- ☐ TR - Turkey
- ☐ TM - Turkmenistan
- ☐ TV - Tuvalu
- ☐ UG - Uganda
- ☐ UA - Ukraine
- ☐ AE - United Arab Emirates
- ☐ GB - United Kingdom
- ☐ US - United States of America
- ☐ UY - Uruguay
- ☐ UZ - Uzbekistan
- ☐ VU - Vanuatu
- ☐ VE - Venezuela
- ☐ VN - Viet Nam
- ☐ YE - Yemen
- ☐ ZM - Zambia
- ☐ ZW - Zimbabwe

4.5 How old are you?

- ☐ 18-30 years
- ☐ 31-55 years
- ☐ +56 years
- ☐ I prefer not to answer

4.6 What is the highest level of education you have completed?

- ☐ Primary or no studies
- ☐ High school
- ☐ Vocational training / Trade school
- ☐ University
- ☐ I prefer not to answer

* 4.7 If you have 15 minutes more, we would be delighted if you could answer some questions about investment decision.

- ☐ Yes, I have some more time.
- ☐ No, I would like to end my contributions here.

5 Investment decisions

In the following questions, you will be asked about specific prices of potential purchases-decisions related to the four scenarios we explained at the beginning of the survey. Please, answer them as honestly as possible consistently with the answer that best fit your lifestyle.

5.1 Appliances:

5.1.1 How much do you think a regular refrigerator costs for personal use?

Only values of at most 20000 are allowed

€

5.1.2 How much do you think an *energy efficient* (e.g., “A” label or similar) refrigerator costs for personal use?

Only values of at most 20000 are allowed

€

5.1.3 If your refrigerator suddenly broke tomorrow, will you buy the energy efficient option?

- ☐ Yes
☐ No
☐ I prefer not to answer

5.1.4 Why no? (please select all that apply to you)

- ☐ a. I do not have enough money for a more an energy efficient one.
☐ b. I do not think buying a more expensive option will solve any problem to the environment or my pocket.
☐ c. I do not think it will be cost-effective to me.
☐ d. I do not trust that energy efficient means high-end, therefore I would rather buy the cheaper option.
☐ e. I do not want to invest money in energy efficient appliances.
☐ f. My peers do not use this kind of appliance.
☐ g. I prefer not to answer.

5.1.5 How much does it have to cost the energy efficient option for you to buy it? (Please answer 0 if you never are willing to pay any extra cost for energy efficiency).

Only values of at most 20000 are allowed

€

5.1.6 Energy efficient refrigerators allow you to save substantial energy (even more than 100€ per year in some cases). This means that in a couple of years the efficient option might in some cases pay its extra costs only with the energy saved. In how many years does the efficient option need to pay back its extra cost for you to invest in it?

Only values of at most 25 are allowed

years

5.2 Insulation:

5.2.1 How much do you think **updating to a state of the art insulation for your façade** (without changing windows) of your house cost? (for multifamily buildings, please think on how much per household costs)

Only values of at most 20000 are allowed

€

5.2.2 If your façade needs to be painted soon, will you invest to update the insulation of your household as well?

- ☐ Yes
☐ No

☐ I prefer not to answer

5.2.3 Why no? (please select all that apply to you)

- ☐ a. I do not have enough money for a more an energy efficient one
- ☐ b. I do not think buying a better insulation will solve any problem to the environment or my pocket.
- ☐ c. I do not think it will be cost-effective to me.
- ☐ d. I do not trust that energy efficient means high-end, therefore I would rather buy the cheaper option.
- ☐ e. I do not want to invest money in insulation.
- ☐ f. My peers have not done this before.
- ☐ g. I prefer not to answer.

5.2.4 How much does it have to cost the energy efficient option for you to buy it? (Please answer 0 if you never are willing to pay any extra cost for energy efficiency).

Only values of at most 20000 are allowed

 €

5.2.5 Insulation allows you to save substantial energy (in some cases more than 80%). This means that in a couple of years the insulation will be amortized only with the energy saved. In how many years does the insulation need to pay back for you to invest in it?

Only values of at most 25 are allowed

 years

5.3 Energy Generation:

5.3.1 How much do you think a **solar photovoltaic system costs**? (for multifamily buildings, please think on how much per household costs)

Only values of at most 20000 are allowed

 €

5.3.2 Are you willing to pay this amount to install a solar photovoltaic system?

- ☐ Yes
- ☐ No
- ☐ I prefer not to answer

5.3.3 Why no? (please select all that apply to you)

- ☐ a. I do not have enough money to install a photovoltaic system.
- ☐ b. I do not think buying a photovoltaic system will solve any problem for the environment or my pocket.
- ☐ c. I do not think it will be cost-effective to me.
- ☐ d. I do not trust that a solar photovoltaic system will provide me any advantage.
- ☐ e. I do not want to invest money in a photovoltaic system.
- ☐ f. My peers have not done this before.
- ☐ g. I prefer not to answer.

5.3.4 How much does the photovoltaic system have to cost for you to invest in it? (Please answer 0 if you never are willing to install it).

Only values of at most 20000 are allowed

 €

5.3.5 Photovoltaic systems allow you to generate your own energy (in some cases more than 60%). This means that in a couple of years the photovoltaic systems will be amortized only with the generated energy. In how many years does the photovoltaic system need to pay back for you to invest in it?

Only values of at most 25 are allowed

 years

5.4 Energy Storage:

5.4.1 How much do you think an **energy storage system** (like batteries) costs? (for multifamily buildings, please think on how much per household costs)

Only values of at most 20000 are allowed

 €

5.4.2 Are you willing to pay this amount to install an energy storage system?

- ☐ Yes
☐ No
☐ I prefer not to answer

5.4.3 Why no? (please select all that apply to you)

- ☐ a. I do not have enough money to install an energy storage system.
☐ b. I do not think buying an energy storage system will solve any problem for the environment or my pocket.
☐ c. I do not think it will be cost-effective to me.
☐ d. I do not trust that an energy storage system will provide me any advantage.
☐ e. I do not want to invest money in energy storage systems.
☐ f. My peers have not done this before.
☐ g. I prefer not to answer.

5.4.4 How much does an energy storage system have to cost for you to invest in it? (Please answer 0 if you never are willing to install it).

Only values of at most 20000 are allowed

 €

5.4.5 Energy storage systems could be used not only to store energy surpluses from photovoltaic solar panels (allowing to generate up to 90% of the energy you consume) but could also provide other revenues (like participating in local flexibility markets, voltage regulation, etc.). This means that in a couple of years the revenues from the energy storage system will have paid its cost. In how many years does the energy storage system need to pay back for you to invest in it?

Only values of at most 25 are allowed

 years

5.5 Heating:

5.5.1 How much do you think a gas boiler costs? (for multifamily buildings with central heating, please think on how much per household costs)

Only values of at most 100000 are allowed

 €

5.5.2 How much do you think an energy efficient heating system (e.g., a heat pump) costs?

Only values of at most 100000 are allowed

 €

5.5.3 If your heating system suddenly brokes tomorrow, will you buy the energy efficient option?

- ☐ Yes
☐ No
☐ I prefer not to answer

5.5.4 Why no? (please select all that apply to you)

- ☐ a. I do not have enough money for a more an energy efficient one
☐ b. I do not think buying an energy efficient heating system will solve any problem for the environment or my pocket.
☐ c. I do not think it will be cost-effective to me.
☐ d. I do not trust that an electric car means high-end, therefore I would rather buy the cheaper option.
☐ e. I do not want to invest money in heating.
☐ f. My peers have not done this before.
☐ g. I prefer not to answer.

5.5.5 How much does it have to cost the energy efficient option for you to buy it? (Please answer 0 if you never are willing to pay any extra cost for energy efficiency).

Only values of at most 100000 are allowed

 €

5.5.6 Heating systems based on heat pumps allow you to save substantial energy. In some cases, heat pumps could use only one fifth of the energy of a similar heating system. This means that in a couple of years the high cost of the heat pump will be amortized only with the energy saved. In how many years does the heat pump need to pay back for you to invest in it?

Only values of at most 25 are allowed

 years

5.6 Mobility:

5.6.1 On average, how many kilometers do you travel in a typical day? Please provide the round trip amount.

Only values of at most 200 are allowed

km

5.6.2 Which travel options do you typically use? Please select all that apply to you

- ☐ a. Private vehicle
- ☐ b. Public Transport
- ☐ c. Walking / Bicycling / Electric Scooter / Other personal mobility devices
- ☐ d. Others
- ☐ e. I prefer not to answer

5.6.3 How much do you think a **diesel or gasoline mid-range brand new car** costs (in €)?

Only values of at most 100000 are allowed

€

5.6.4 How much do you think a **mid-range brand new electric car** costs?

Only values of at most 100000 are allowed

€

5.6.5 If your car suddenly broke, are you willing to buy an electric car?

- ☐ Yes
- ☐ No
- ☐ I prefer not to answer

5.6.6 Why no? (please select all that apply to you)

- ☐ a. I do not have enough money to buy an electric car.
- ☐ b. I do not think buying an electric car will solve any problem for the environment or my pocket.
- ☐ c. I do not think it will be cost-effective to me.
- ☐ d. I do not trust that an electric car means high-end, therefore I would rather buy the cheaper option.
- ☐ e. I do not want to invest money in electric cars.
- ☐ f. My peers have not done this before.
- ☐ g. I prefer not to answer.

5.6.7 How much does an electric car have to cost for you to invest in it? (Please answer 0 if you never are willing to pay any extra cost for a car).

Only values of at most 100000 are allowed

€

5.6.8 Electric cars allow you to save substantial energy (even more than 2000€ per year in some cases). This means that in a couple of years the electric car will pay it extra costs only with the energy saved. In how many years does the electric car need to pay back its extra cost for you to invest in it?

Only values of at most 25 are allowed

years

6 Household Characterization

6.1 Please select zero

- ☐ Zero
- ☐ Two
- ☐ Ten
- ☐ Six

6.2 What type of area do you live in?

- ☐ Rural
- ☐ City
- ☐ Residential area
- ☐ I prefer not to answer

6.3 Which of the following most accurately describes you?

- ☐ Male
- ☐ Female
- ☐ Non-binary
- ☐ Other
- ☐ I prefer not to answer

6.4 Do you live alone or do you share your household with others? (Please check all the answers that apply to you and your personal situation)

- ☐ a. I live alone.
- ☐ b. I live with friends or other non-family persons.
- ☐ c. I live with my partner.
- ☐ d. I live with my child/children (or other minor/s under my responsibility).
- ☐ e. I live with other adult/s under my responsibility.
- ☐ f. I live with other person/s who are either family or relative/s.
- ☐ g. I live with my pet/s.
- ☐ h. I prefer not to answer.

6.5 How many people live in the household?

Only values of at most 20 are allowed

6.6 What is your approximate individual yearly income (net)?

- ☐ More than 100.000 €
- ☐ Between 50.000 € and 100.000 €
- ☐ Between 30.000 € and 50.000 €
- ☐ Between 15.000 € and 30.000 €
- ☐ Less than 15.000 €
- ☐ I prefer not to answer

6.7 In a typical month, how much of your salary is saved?

- ☐ I manage to save more than 20% of the monthly salary.
- ☐ I manage to save between 10 and 20 % monthly.
- ☐ I manage to save less than 10 % monthly.

- ☐ Nothing, I am using my savings lately.
- ☐ I prefer not to answer.

6.8 In which type of housing do you live?

- ☐ Flat / Apartment
- ☐ Semi-Detached / Townhomes
- ☐ Single-Family Home / Fully detached house
- ☐ In a multi-family house
- ☐ Other type of household
- ☐ I prefer not to answer

6.9 Are you the owner / co-owner of the place where you live?

- ☐ Yes
- ☐ No
- ☐ I prefer not to answer

6.10 How much usable area does your house have?

- ☐ Less than 80 m²
- ☐ Between 80 and 120 m²
- ☐ More than 120 m²
- ☐ Not sure
- ☐ I prefer not to answer

6.11 Is your house properly insulated?

- ☐ Yes - it has been built or refurbished with energy efficiency in mind.
- ☐ No - it has been built without energy efficiency in mind or has not been completely refurbished.
- ☐ I do not know.
- ☐ I prefer not to answer.

6.12 In which climate zone do you live?

- ☐ Atlantic
- ☐ Mediterranean
- ☐ Continental
- ☐ Alpine or Nordic
- ☐ Arid or semi-arid
- ☐ I do not know
- ☐ I prefer not to answer

6.13 On a scale of 0-10, where 0 is "Climate change does not exist" and 10 is "I am a climate change expert /activist", how would you assess your level of awareness of climate change?

6.14 On a scale of 0-10, where 0 is "Climate change does not exist" and 10 is "They are climate change experts/activists", how would you assess the level of awareness of climate change of **your peers or relatives**?

6.15 On a scale of 0-10, where 0 is “It is the first time I hear about it” and 10 is “I am an expert or activist”, how would you assess your level of awareness about the energy transition?

6.16 On a scale of 0-10, where 0 is “It is the first time they hear about it” and 10 is “They are experts or activists”, how would you assess the level of awareness about the energy transition of **your peers or relatives**?

6.17 Please provide below any additional feedback about the survey you could give us.